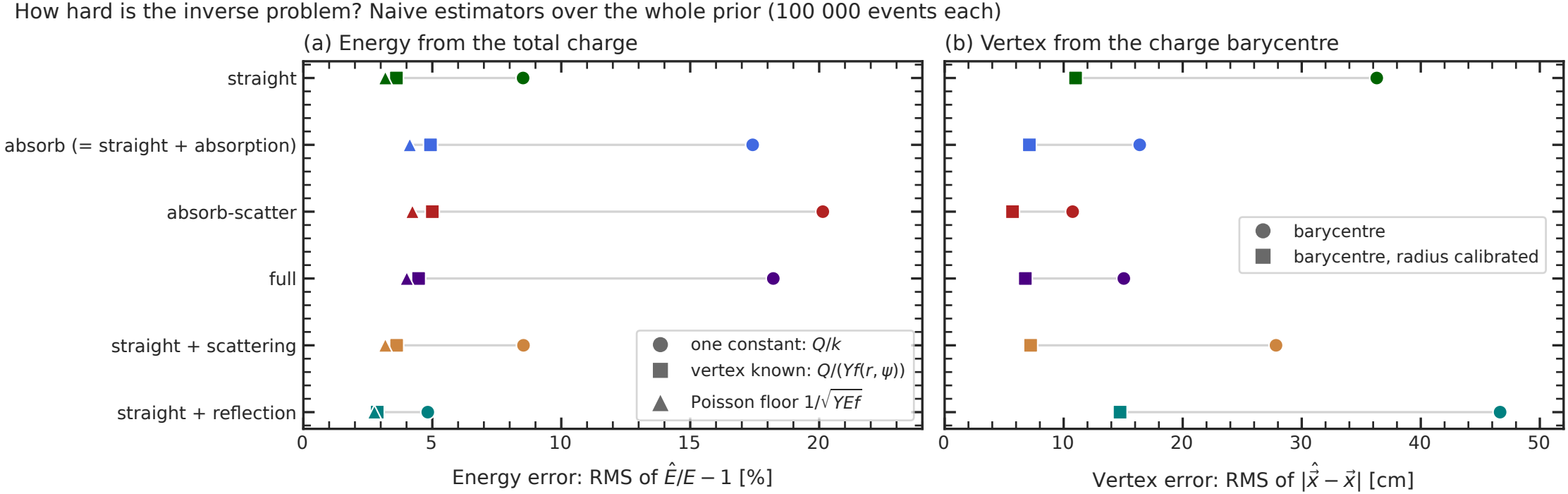
How hard is the inverse problem? Naive estimators over the whole prior (100 000 events each)
(a) Energy from the total charge
(b) Vertex from the charge barycentre
straight
absorb (= straight + absorption)
absorb-scatter
full
straight + scattering
straight + reflection
one constant: Q/k
vertex known: $Q/(Yf(r,\psi))$
Poisson floor $1/\sqrt{YEf}$
barycentre
barycentre, radius calibrated
0
5
10
15
20
30
40
50
Energy error: RMS of $\hat{E}/E - 1$ [%]
Vertex error: RMS of $|\hat{\vec{x}} - \vec{x}|$ [cm]